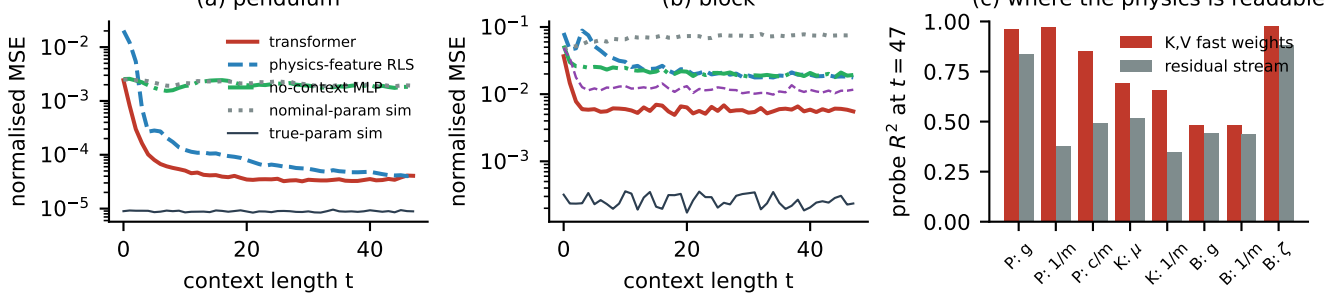
(a) pendulum
normalised MSE
10^{-2}
10^{-3}
10^{-4}
10^{-5}
transformer
physics-feature RLS
no-context MLP
nominal-param sim
true-param sim
0
20
40
context length t
(b) block
normalised MSE
10^{-1}
10^{-2}
10^{-3}
0
20
40
context length t
(c) where the physics is readable
probe R^2 at $t = 47$
1.00
0.75
0.50
0.25
0.00
K,V fast weights
residual stream
P: g
P: 1/m
P: c/m
K: μ
K: 1/m
B: g
B: 1/m
B: ζ